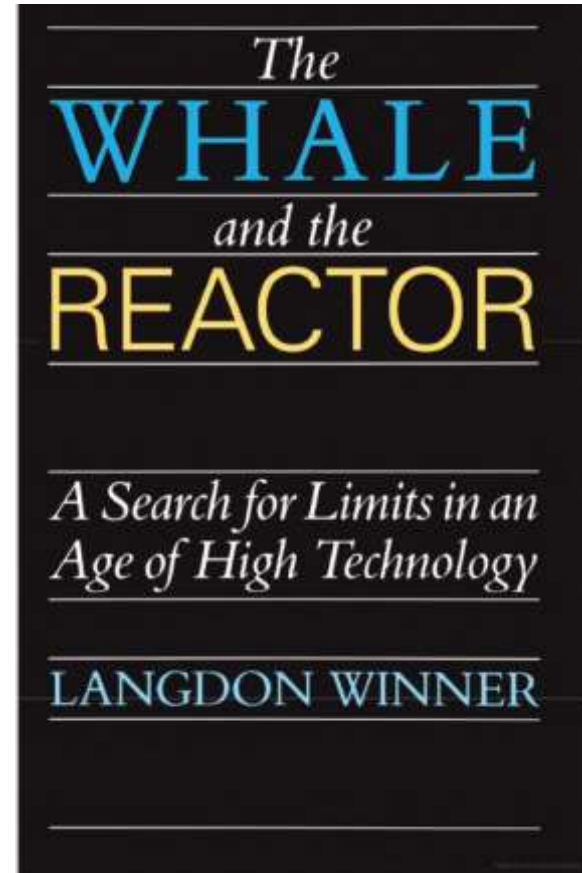


Sociologia Digital e IA
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The Whale and the Reactor

Technology as a form of life
&
Do artifacts have politics?



Preface

For most people it is enough to know how technical systems are produced, how they are run, how they are best used, and how they contribute to that vast aggregate of blessings: economic growth (p.ix).

My aim here is to go further, exploring the meaning of technology for the way we live (p.ix-x).

Preface

How can one look beyond the obvious facts of instrumentality to study the politics of technical objects (p.x)?

Preface

A search for limits. In an age in which the inexhaustible power of scientific technology makes all things possible, it remains to be seen where we will draw the line, where we will be able to say, here are the possibilities that wisdom suggests we avoid (p.xi).

Technological Somnambulism

[...] engineers appear unaware of any philosophical questions their work might entail (p.4).

[...] it is usually taken for granted that the only reliable sources for improving the human condition stem from new machines techniques and chemicals (p.5)

Technological Somnambulism

[...] technology has never joined epistemology, metaphysics, esthetics, law, science and politics as a fully respectable topic for philosophical enquiry (p.4).

Technological Somnambulism

Technologies are powerful forces acting to reshape human activity and its meaning (p.6).

Thus, the deceptively obvious division between the making and the use of tools is insufficient as an analytical framework.

Once again, the question concerning the neutrality of technology.

Technological Somnambulism

Technological devices tend to engender worlds of their own: the case of the automobile (p.8).

In cities like Los Angeles, the simple act of walking can be cause for alarm (p.9).

Technological Somnambulism

In the technical realm we repeatedly enter into social contracts, the terms of which are revealed only after the signing (p.9).

The interesting puzzle of our times is that we so willingly sleepwalk through the process of reconstituting the conditions of human existence (p. 10).

Beyond Impacts and Side Effects

After the Bulldozer has rolled over us, we can pick ourselves up and carefully measure the treadmarks (p.10).

It is often assumed that those who experience the effects of technological change are obliged simply to endure it (p.10).

Beyond Impacts and Side Effects

The construction of a technical system that involves human beings as operating parts (sic) brings a reconstruction of social roles and relationships (p.11).

We carry with us highly structured anticipations about entities that appear to participate, if only minimally, in forms of life and associated language games that are parts of human culture (p.14).

Beyond Impacts and Side Effects

[...] the philosophies of Marx and Wittgenstein direct our attention to the fabric of everyday existence.

Some remarks on Marx's broad concept of production and its bearings to the philosophy (and sociology) of technology (p.14-15).

Beyond Impacts and Side Effects

It makes sense to ask, for example, how the adoption of digital computers might alter the way people think of their own faculties and activities (p.15).

Return to Making

[...] the philosophies of Marx and Wittgenstein share a fruitful insight: the observation that social activity is an ongoing process of world-making.

[but...]

Marxist theory upholds an attitude as nearly as lethargic as the Wittgensteinian decision to 'leave everything as it is' (p.17).

Return to Making

As we 'make things work', what kind of world are we making?

[...] we must admit our responsibility for what we are making (p.18).

“we”: who? Under what circumstances? What has been the role of computer systems in recent decades?

On the idea that artifacts have politics

Lewis Mumford on '*democratic, human-centered x authoritarian, systems-centered technologies*': the theory and its pitfalls (p.19).

Example: arguments over nuclear x solar power.

On the idea that artifacts have politics

Scarcely a new invention comes along that someone doesn't proclaim it as the salvation of a *free society* (p.20, grifo nosso).

On the idea that artifacts have politics

Common 'advice': what matters is not technology itself, but the social or economic system in which it is embedded (p.20).

Social determination of technology as an important counterpoint to technological determinism (albeit insufficient): the point is that things matter.

On the idea that artifacts have politics

The theory of technological politics draws attention to the momentum of large-scale sociotechnical systems, the response of modern societies to certain technological imperatives, and to the ways human ends are powerfully transformed as they are adapted to technical means (p.21).

On the idea that artifacts have politics

The term 'technology' is understood to mean a modern practical artifice [...], but I prefer to speak of 'technologies' plural, smaller or larger pieces or systems of hardware of a specific kind (p.22).

Pieces of hardware and software, would be a best take.

Technical arrangements and social order

[...] some two hundred or so low-hanging overpasses on Long Island are there for a reason [...]

[...] generations after Moses' death and the alliances he forged have fallen apart, his public works [...] will continue to shape [New York City]

“Moses is regarded as one of the most powerful and most influential people in the history of New York City and New York state” (wikipedia).

Technical arrangements and social order

If we suppose that new technologies are introduced to achieve increased efficiency, the history of technology shows that we will sometimes be disappointed (p.24).

Technical arrangements and social order

It is obvious that technologies can be used in ways that enhance the power, authority and privilege of some over others [...] (p.25).

Technical arrangements and social order

Some technologies encompass purposes far beyond their immediate use (p.25).

We must be attentive to the meaning of the designs and arrangements of our artifacts.

Technical arrangements and social order

The tomato harvest case:

An ongoing social process in which scientific knowledge, technological invention and corporate profit reinforce each other in deeply entrenched patterns, patterns that bear the unmistakable stamp of political and economic power (p.27).

Technical arrangements and social order

Ranges of choices concerning technological innovation: (1) Whether to build or not to build something; (2) Define specific features in the design or arrangement of a technical system (p.28).

To the “tangible arrangements of steel and concrete, wires and semiconductors, nuts and bolts” we must add algorithms and data.

Inherently political technologies

Upon approaching Engels' and Plato's arguments about the relationship between authority and technical systems, Winner seems to miss the point, by confusing authority with authoritarianism (p.30-31)...

Inherently political technologies

Nevertheless, he raises an important question:
What does modern technology make possible or necessary in political life (p.32)?

On this matter, Winner's distinction between "internal" and "external" social relationships remain unclear.

Critical assessment

A somewhat loose reading of both Marx and Wittgenstein.

A seemingly lack of hands-on experience (or even connoisseurship) at sociotechnical settings such as manufacturing and software development.

Unclear concepts of freedom, authority and authoritarianism.

Further developments

It remains to be discussed to what extent technologies are necessary or contingent (the electronic digital computer is a case in point).